

1. Business Problem Statement

Food Delivery Analytics Project

1.1 Business Challenge

The business needs to understand delivery-time variation and the operational conditions associated with slower deliveries, then turn the analysis into decision-ready performance information.

- 1 Identify the key factors contributing to longer delivery times.
- 2 Compare delivery performance across cities.
- 3 Compare delivery performance across order hours.
- 4 Understand the role of traffic, weather, distance, preparation time and multiple deliveries.
- 5 Create reliable KPIs for management monitoring and operational decisions.

1. Dataset

41,953 initial records

2. KPI

26.34 min initial mean

3. Missing Time

4,133 records

4. Columns

17 initial columns

2. Problem Scope

What the analysis is intended to explain

2.1 Core Performance Question

The notebook treats `Time_Taken_Min` as the central delivery-performance measure and uses descriptive analysis to understand how it behaves across operational dimensions.

- ① Delivery-time distribution shows the overall pattern of delivery duration.
- ② City-level averages identify geographic differences in average delivery time.
- ③ Order-hour averages identify time-of-day differences.
- ④ Festival vs non-festival order counts provide demand-context information.
- ⑤ Correlation analysis explores relationships among numeric variables.

2.2 Business Need

A single average is not enough. Management needs location and time views plus operational context to identify where attention is required.

3. Data Quality & Preparation

Why reliable preparation is part of the business problem

- ❶ Columns with more than 50% missing values are removed.
- ❷ Order_Date is converted to datetime with errors coerced.
- ❸ Numerical missing values are filled using the median.
- ❹ Categorical missing values are filled using the mode.
- ❺ Time_Taken_Min is converted to float after preparation.
- ❻ Duplicate records are checked.
- ❼ IQR is used to define lower and upper limits for Time_Taken_Min.
- ❽ Delivery-time records outside the IQR limits are filtered.

1. Cleaning

Median + mode

2. Outliers

IQR method

3. Validation

Duplicate check

4. Output

Excel analytics file

4. Analytical Business Questions

Questions the completed workflow can answer

1

What is the overall distribution of delivery time?

2

Which cities have the highest average delivery time?

3

How does average delivery time change by order hour?

4

What is the split between festival and non-festival orders?

5

Which numeric variables move together with delivery time?

Decision focus: identify patterns that can guide operational review and dashboard monitoring.

5. Business Success Criteria

How the problem statement translates into measurable deliverables

1

Reliable delivery-time KPI

Use a cleaned and consistently prepared Time_Taken_Min measure.

2

Location visibility

Provide city-level average delivery-time comparison.

3

Time visibility

Provide order-hour delivery-time trend.

4

Operational context

Use available traffic, weather, distance and delivery-related variables for interpretation.

5

Management usability

Present findings in a dashboard structure that supports recurring decisions.

Business goal: reduce delays and make delivery performance more predictable.